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	Document Title:	RESEARCH PROPOSAL	Revision No.	00
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(1) Research Program/Project/Study Title:	Library Mobile Application for MSC
Proponent Name:	Nicolas V. Braña IV
Academic Rank:	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Mambusao
		Province:	Capiz
		Region:	6
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus)	

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☐ Collaborative Research (external agency)	
(7) Sector/Commodity	(8) Discipline/Field of Specialization Information Technology
(9) Rationale The Library Mobile Application is a comprehensive solution designed to provide up-to-date information on the availability of their books. This system improve the library services, and optimize time uses of the faculty and the students in browsing the shelves of library within CAPSU Mambusao Satellite College. The Library Mobile Application is to ensure the accurate number and list of books available in the library. Manual searching for books methods can be time consuming specially for the faculty of CAPSU Mambusao Satellite College. By Using the Library Mobile Application, the system provides the list of the books to be browse, minimizing the time consumed in manual browsing of books in the CAPSU Mambusao Satellite College library.	
(9-A) Project Description The Library Mobile Application is a valuable for CAPSU Mambusao Satellite College Students and Faculty to streamline or browse and provide knowledge about the books availability. By developing and implementing's this system, CAPSU Mambusao Satellite College can ensure accurate number of books, leading to better time utilization and improved book browsing in CAPSU Mambusao Satellite College. Mobile Application: Designed to provide up-to-date information. Time Management: The system provides a faster way to browse the available books in CAPSU Mambusao Satellite College Library	
(10) Objectives/Statement of the Problem 1. Create a Library Mobile Application for CAPSU Mambusao Satellite College Students and Faculty. 2. Design an interface for accurate list of books. 3. Develop a mobile application that can generate real-time reports and provide the list of books, and a user-friendly interface for students and faculty of CAPSU Mambusao Satellite College. 4. Evaluate a system using ISO 25010 standard (in terms of appearance, design, efficiency, and its operating performance.	
(11) Review of Literature/Prior Art Search Library book reservation method based on ultrahigh-frequency RFID technology by Lijun Chen, Yuanjie LOU, Qingpu SHI, Linghao WAN, Ying Li, Xi Chen, Afan ZENG, Bin Wang Objective of the invention: the technical problem to be solved by the present invention is to provide a library book reservation method based on ultra high frequency RFID technology in view of the shortcomings in the prior art. The arrangement sequence of each layer of books in a library is automatically identified by using a radio frequency phase in the ultra high frequency RFID technology. In order to solve the technical problems, the present invention discloses a library book reservation method based on ultra high frequency RFID technology, including a method for eliminating books not in the present layer on a bookshelf based on the change rate of a radio frequency phase, and a method for sorting books based on a time-varying change rule of the radio frequency phase. On our research we will use the database in the library system to use in the mobile application, to accurately view the exact amount and books in the library by this it will minimize the time consume in searching for books in the library shelves. Display method and device for collected application program page by Tang Yong The invention provides a method for displaying collected contents, which comprises the following steps: responding to a page information collection instruction of a current page of a second application running in the foreground by a first application running in the background, and acquiring a uniform resource locator of the page; adding a uniform resource	

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locator of the page into a favorite list of the first application; and outputting a suspension piece providing an access entrance of the page information pointed by the uniform resource locator in the favorite list at a terminal user interface. The method can provide an access entry for application contents collected by different application programs, and the access entry is represented and output on a terminal interface in the form of a suspension piece, so that page information pointed by the uniform resource locator can be quickly entered from the suspension piece, the first application and other applications can be quickly switched, convenience and quickness are realized, the collection and access efficiency is improved, and the use experience of a user is improved. On our research we will use the collected information from the library books by their code, category, and the type of books. The method can provide an access to the mobile applications services.

Techniques for providing retail customers a seamless, individualized discovery and shopping experience between online and physical retail locations by William G. Crutchfield Jr.

The present invention is directed to techniques for providing retail customers a seamless, individualized discovery and shopping experience. With an improved shopper/customer accounting system, along with other technologies such as intelligent fixtures, mobile devices, and product simulations, a shopper's product discovery, purchase, and post-sale experience can be highly integrated and personalized. For example, the shopper's out-of-store (online) and in-store product discovery sessions may be seamlessly linked and analyzed to provide the shopper with customized recommendations and assistance. Other variations and enhancements are disclosed.

Similarly to the system where the application can provide the location and the category of the book needed by the customer

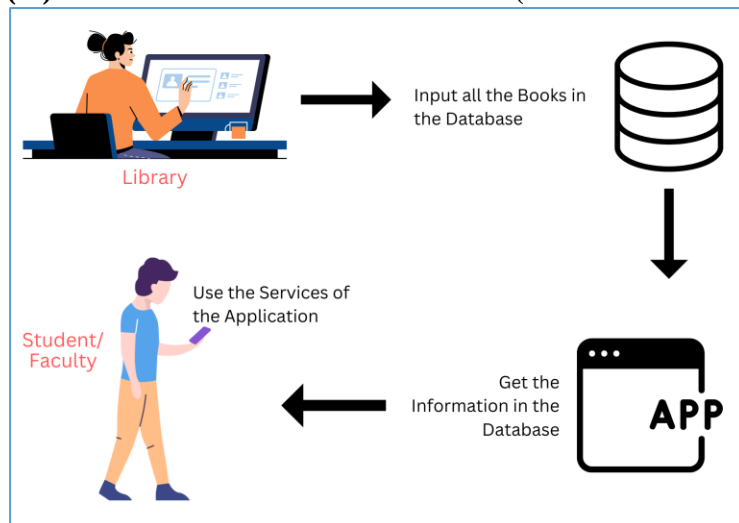
Mobile media pause and resume by Kavi Maharajh, Bart MacNeil, Torin Walker

In embodiments, the present invention provides a method and system for managing playback of content delivered to a mobile device with a pause and resume functionality. The method and system including receiving delivered content on a mobile device, initiating playback of the delivered content, pausing playback of the content being delivered to the mobile device in response to an action and resuming delivery of the content upon a request. On our Application the application have a realtime function to ensure the accurate services.

Systems and methods for virtual markets with product pickup by Ginger Casey

A system and method for virtual markets with product pickup generally comprised of a three-dimensional user supermarket interface, where a user places an order, and a retail/warehouse hybrid procurement center, comprised of a fulfillment warehouse and a pickup bay, where the order is generally fulfilled, packaged and picked-up by a user. Similarly to our application where you can browse the list of books and their availability.

(12) Scientific Basis/Theoretical Framework (



(13) Methodology

The Software Development Life Cycle (SDLC) for the methodology in developing the CAPSU Library Mobile Application

Project Planning (Requirements Gathering)

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Identify and document the specific requirements and objectives of the system based on CAPSU's needs assessment. The researchers will gather all the information of the books in the library that is compatible in the mobile application

Designing (System Design)

Design the system architecture, user interface, and database structure based on the identified requirements. Define the functionalities and features of the system. The researchers will use the database in the library for the list of books, to provide an exact and accurate list of books to be browsed in the library mobile application.

Coding (System Development)

Develop the Library Mobile Application for MSC based on the design specifications. Follow coding best practices, maintain coding standards, and ensure the system is modular, scalable, and maintainable. Conduct regular code reviews and testing to identify and fix any bugs or issues during development.

Testing (System Testing)

Perform different types of testing, such as unit testing, integration testing, system testing, and user acceptance testing. Validate that the system meets the defined requirements and functions as expected. Identify and resolve any defects or issues discovered during testing.

Deployment (Releasing)

Prepare the system for deployment by configuring it according to CAPSU's infrastructure and requirements.
 Plan the deployment strategy, including any necessary hardware or software installations.
 Conduct a pilot deployment in a controlled environment to ensure system stability and compatibility.

By following this SDLC methodology, the purposed of the Library Mobile Application for MSC will be carried out systematically, ensuring a structured approach to development, testing, deployment, and ongoing maintenance.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1						Year 2					
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
1. Planning												
2. Designing												
3. Coding												
4. Testing												
5. Deployment												

(15) Expected Output

A user-friendly interface mobile application
 Can provide the accurate list of books for Students and faculty.
 A mobile application that can generate real-time reports

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

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(17) Target Beneficiaries

CAPSU Mambusao Satellite College Students and Faculty will be able to browse the library using mobile phones.

(18) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study:

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		15,000
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		50,000
- Transportation allowance		5,000
- Meal allowance during data gathering		3,000
- Professional fees (laboratory analysis and other related fees)		40,000
- Miscellaneous (for incidental expenses)		10,000
- Printing		10,000
C. CAPITAL OUTLAY		133,000
Total		

(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

<https://patents.google.com/patent/US11004017B2/en?inventor=Lijun+Chen&oq=Lijun+Chen>

[https://patents.google.com/patent/CN107092707B/en?q=\(Display+method+and+device+for+collected+application+program+page\)&oq=Display+method+and+device+for+collected+application+program+page+](https://patents.google.com/patent/CN107092707B/en?q=(Display+method+and+device+for+collected+application+program+page)&oq=Display+method+and+device+for+collected+application+program+page+)

<https://patents.google.com/patent/US10977701B2/en?inventor=William+G.+Crutchfield+Jr.&oq=William+G.+Crutchfield+Jr.>

[https://patents.google.com/patent/US11675836B2/en?q=\(Mobile+media+pause+and+resume\)&oq=Mobile+media+pause+and+resume+](https://patents.google.com/patent/US11675836B2/en?q=(Mobile+media+pause+and+resume)&oq=Mobile+media+pause+and+resume+)

[https://patents.google.com/patent/US20210133837A1/en?q=\(Systems+and+methods+for+virtual+markets+product+pickup\)&oq=Systems+and+methods+for+virtual+markets+with+product+pickup+](https://patents.google.com/patent/US20210133837A1/en?q=(Systems+and+methods+for+virtual+markets+product+pickup)&oq=Systems+and+methods+for+virtual+markets+with+product+pickup+)

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(21) Capsule Curriculum Vitae

NICOLAS V. BRAÑA IV

FACULTY

 Capiz State University Mambusao Satellite College
 09776354282
 nvbrana4@capsu.edu.ph



EDUCATIONAL BACKGROUND

Full Name : Nicolas Villas Braña IV
 Gender : Male
 Date of Birth : January 8, 1996
 Age : 28
 Civil Status : Single

CAPIZ STATE UNIVERSITY MAMBUSAO SATELLITE COLLEGE

Bachelor of Science in
 Computer Science
 2018 - 2019

CAPIZ STATE UNIVERSITY MAMBUSAO SATELLITE COLLEGE

Computer Programming NCIV
 2018 - 2019

CENTRAL PHILIPPINE UNIVERSITY

Bachelor of Science in Packaging
 Engineering(Undergrad)
 2012 - 2015

EXPERIENCE

Faculty

2022 - Present

- Capiz State University Mambusao Satellite College

Administrative Staff

2021 - 2022

- Capiz State University Research, Development and Extension Office

Cashier/Foodservice(FS).

2019 - 2020

- PRINCEWORLD MARKETING (I-Mart Mambusao)

Skills

- Flexible Adaptability
- Computer Literate
- Critical Thinking
- Proficient in using (Microsoft Word, Microsoft Excel, Microsoft PowerPoint and Adobe Photoshop)